📌 Software Testing Principles

1. Testing Shows Presence of Defects

Testing can show bugs exist, but cannot prove the software is completely bug-free.

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2. Exhaustive Testing is Impossible

Testing every possible input and scenario is impossible.

Example:
A login form may have millions of combinations.

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3. Early Testing Saves Cost

Finding bugs early reduces fixing cost and effort.

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4. Defect Clustering

Most bugs are usually found in a small number of modules.

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5. Pesticide Paradox

Running the same test cases repeatedly may stop finding new bugs.

Solution:
Regularly update test cases.

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6. Testing Depends on Context

Testing approach changes depending on the application type.

Example:
Banking system testing differs from gaming application testing.

🧩 Levels of Testing

🔹 Unit Testing

Testing individual components or functions.

Usually done by developers.

🔹 Integration Testing

Testing interaction between modules.

Example:
Login module communicating with database.

🔹 System Testing

Testing the complete application as a whole.

🔹 Acceptance Testing

Testing whether the system satisfies business requirements.

Types

- Alpha Testing
- Beta Testing

🌐 Black Box Testing

Tester checks functionality without seeing internal code.

Focuses On

- Inputs
- Outputs
- User behavior

Example:
Testing login functionality without knowing backend code.

⚪ White Box Testing

Testing internal code structure and logic.

Usually performed by developers.

⚫ Gray Box Testing

Combination of Black Box and White Box testing.

Tester has partial knowledge of internal system.

📱 Compatibility Testing

Checks whether application works across:

- Different browsers
- Different devices
- Different operating systems

Example

- Chrome
- Firefox
- Edge
- Mobile devices

🔒 Security Testing

Checks application security.

Examples

- Password protection
- Unauthorized access prevention
- Secure data handling

⚡ Performance Testing

Checks system speed and stability.

Includes

- Load Testing
- Stress Testing
- Scalability Testing

🗂️ Requirement Traceability Matrix (RTM)

A document used to map requirements with test cases.

Purpose:

- Ensure all requirements are tested
- Track testing coverage

📝 Bug Report Format

A professional bug report contains:

Field	Example
Bug ID	BUG_001
Title	Login button not working
Description	User cannot login with valid credentials
Steps to Reproduce	Enter valid username and password
Expected Result	User logs in
Actual Result	Login fails
Severity	High

Field	Example
Priority	High
Status	Open

📌 Entry Criteria & Exit Criteria

Entry Criteria

Conditions required before testing starts.

Example

- Requirements available
- Build deployed
- Test environment ready

Exit Criteria

Conditions required before testing ends.

Example

- All critical bugs fixed
- Test cases executed
- Test summary completed

🧪 Test Environment

Setup used for testing software.

Includes:

- Hardware
- Software
- Network

- Database
- Testing tools

📄 Test Summary Report

Document prepared after testing completion.

Contains:

- Number of test cases executed
- Passed/failed count
- Bug summary
- Testing status

🛠️ QA Roles and Responsibilities

A QA Engineer typically:

- Understands requirements
- Writes test cases
- Executes tests
- Reports bugs
- Performs regression testing
- Prepares test reports

📌 Agile Testing

Testing performed in Agile software development.

Characteristics

- Continuous testing
- Fast feedback
- Team collaboration

- Sprint-based work

🔄 Waterfall vs Agile

Waterfall	Agile
Sequential process	Iterative process
Testing after development	Continuous testing
Less flexibility	More flexibility

🧪 API Testing Basics

Testing backend APIs without UI.

Common Checks

- Status codes
- Response body
- Response time
- Authentication

Popular Tool

- Postman

🌍 Browser Testing

Manual testers commonly test on:

- Google Chrome
- Mozilla Firefox
- Microsoft Edge

📱 Mobile Testing

Testing mobile applications on:

- Android
- iOS
- Tablets

Checks

- Responsive design
- Screen rotation
- Notifications
- App installation

📊 Common QA Metrics

Examples:

- Test Case Pass Percentage
- Defect Density
- Defect Leakage
- Test Coverage

💡 Tips for Beginner QA Engineers

- Practice writing test cases daily
- Learn bug reporting clearly
- Use demo applications
- Understand SDLC and STLC well
- Learn API testing basics
- Build GitHub QA projects
- Practice real-world scenarios

🎯 Sample Manual Testing Project Ideas

1. Login Page Testing

Test login validations and authentication.

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2. E-commerce Website Testing

Test:

- Add to cart
- Checkout
- Search
- Payment flow

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3. Banking Application Testing

Test:

- Money transfer
- Transaction history
- Login security

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4. Student Management System Testing

Test:

- Student registration
- Login
- Attendance
- Results

🎤 Additional Interview Questions

What is STLC?

Software Testing Life Cycle — process followed during software testing.

What is a Bug Life Cycle?

The stages a bug goes through from identification to closure.

What is Compatibility Testing?

Testing application behavior across different browsers/devices/platforms.

What is Retesting?

Testing fixed defects again to verify fixes.

Difference between Retesting and Regression Testing?

Retesting	Regression Testing
Checks fixed bug	Checks entire application
Specific functionality	Related functionalities

🚀 Useful Practice Sites

- OrangeHRM Demo
- Sauce Demo
- Demo Web Shop
- Automation Exercise